STRAVA FITNESS DATA ANALYSIS

Tableau Dashboard & Insights

Project	Strava Fitness Analytics
Tool Used	Tableau Public
Records Analysed	940 activity records
Analysis Period	12 April 2016 – 12 May 2016

This report gives a short overview of the Tableau dashboard created for the Strava fitness dataset. The dashboard was used to look at activity, steps, calories, active minutes and sleep patterns.

1. Objective and Dataset Overview

The main aim of the Tableau analysis was to understand how activity levels, daily steps, active minutes and sleep duration are distributed in the dataset. The dashboard was also designed to make the results easier to compare using charts and filters.

Main fields used

Field	Used for
Activity Date	Daily trend and date filtering
Total Steps	Daily steps and activity-level comparison
Calories	Average daily calorie KPI
Very Active Minutes	Very active time KPI
Fairly / Lightly / Sedentary Minutes	Activity intensity analysis
Activity Level	Low, Moderate and High activity comparison
Sleep Segment	Sleep duration vs average steps

Dashboard KPIs

940	Total Activity Records
7,638	Average Daily Steps
2,304	Average Daily Calories
21.16	Average Very Active Minutes

2. Tableau Dashboard

The final Tableau dashboard brings the main KPIs and four charts together on one page. An Activity Date filter and an Activity Level filter were added so that the dashboard can be explored interactively.

940 **Total Activity Records**	**7,638** **AVG DAILY STEPS**
2,304 **AVERAGE DAILY CALORIES**	**21.16** **AVERAGE VERY ACTIVE MINUTES**

Daily Steps Trend	**Steps by Activity Level**
Activity Minutes by Intensity Segment	**Sleep Duration vs Average Daily Steps**

The dashboard layout keeps the KPIs at the top, followed by the daily trend and activity comparisons. This makes the main numbers visible first and the detailed charts easier to read below.

3. Key Findings from the Dashboard

Daily Steps Trend

Daily total steps stayed mostly in the roughly 200K–270K range during most of the period. There are several day-to-day fluctuations, with a noticeable drop at the end of the period on 12 May.

Activity Level

The High Activity group has the highest average total steps, followed by Moderate Activity and then Low Activity. This follows the expected pattern in the dataset and shows that the activity classification separates the groups clearly.

Activity Intensity

The low very-active intensity segment has the largest total amount of activity minutes in the displayed chart. The moderate and high very-active segments have smaller totals.

Sleep and Steps

The displayed averages are 9,484 steps for the Less than 6 Hours group, 8,791 for the 6–8 Hours group and 7,000 for the More than 8 Hours group. This is an observed pattern in this dataset; it should not be treated as proof that less sleep causes more steps.

Filters Used

The dashboard includes Activity Date and Activity Level filters. These allow the user to focus on a particular date range or activity category without rebuilding the charts.

4. Conclusion

The Tableau dashboard provides a simple view of the main fitness patterns in the Strava dataset. The KPIs give a quick summary of the data, while the charts help compare activity levels, intensity and sleep segments.

From the analysis, higher activity levels are associated with higher average step counts in the displayed data. Daily activity also changes from day to day, and the sleep segmentation shows different average step levels across the three groups.

Tools Used

• Tableau Public for dashboard creation and visualization

• Filters for interactive exploration of Activity Date and Activity Level

• Bar charts and line chart for comparison and trend analysis

Overall, the dashboard converts the raw fitness data into a few easy-to-understand visual insights and provides an interactive way to explore the activity patterns.

— End of Tableau Analysis —